

Explainable Diabetic Retinopathy Grading on Consumer Hardware: Quantised Low-Rank Adaptation of a 2B Vision-Language Model

Yuvraj Verma 

Independent Researcher

Email: yuvrajverma282004@gmail.com

ORCID: 0009-0004-2138-3159

Abstract—Diabetic retinopathy (DR) is a leading cause of preventable blindness, and automated grading is widely proposed as a means of extending screening capacity. Most deployed systems, however, are opaque classifiers that emit a severity label without justification, while the vision-language models (VLMs) capable of articulating their reasoning are generally assumed to require data-centre accelerators for adaptation. This paper examines whether clinically useful, explainable DR grading can instead be obtained on a single consumer GPU. We construct IDRiD-VQA, a visual-question-answering corpus of 2,064 label-grounded question-answer pairs spanning grade identification, lesion identification, clinical reasoning and patient recommendation, derived from the 516 fundus photographs of the IDRiD disease-grading subset. We then adapt Qwen2-VL-2B-Instruct using 4-bit quantised low-rank adaptation, training only 28.9M parameters (1.45% of the model) within a 6GB memory envelope. On the canonical 103-image test split, adaptation raises exact-match grading accuracy from 31.1% to 66.0% and quadratic weighted kappa from 0.00 to 0.76, while roughly tripling ROUGE-L agreement of the generated explanations and reducing inference latency by 47%. A controlled ablation isolates input resolution as the dominant factor: increasing the shorter training resolution from 512 to 768 pixels contributes 11.6 accuracy points and 0.14 kappa at negligible memory cost, because aggressive downsampling destroys the microaneurysms that define early disease. The complete fine-tune requires approximately 90 minutes and peaks at 3.52 GB of VRAM. A per-class analysis, however, qualifies these aggregate figures sharply: the adapted model collapses onto the two majority grades and never predicts mild retinopathy, so accuracy and ordinal agreement conceal a complete failure at the stage where screening is most actionable. We release the dataset, adapter weights and code to support reproducible low-resource medical VLM research, and we argue that aggregate agreement should not be mistaken for clinical readiness.

Index Terms—diabetic retinopathy, vision-language models, parameter-efficient fine-tuning, QLoRA, explainable artificial intelligence, medical image analysis

I. Introduction

Diabetic retinopathy is a microvascular complication of diabetes mellitus and one of the principal causes of avoidable vision loss worldwide; recent meta-analytic estimates place global prevalence among people with diabetes near

one in five, with the absolute burden projected to grow substantially through 2045 [1]. Screening programmes rely on the interpretation of colour fundus photographs against the International Clinical Diabetic Retinopathy scale, which partitions severity into five ordinal grades [2]. The clinical bottleneck is not imaging but interpretation: the supply of trained graders is far smaller than the population requiring periodic examination.

Automated grading has therefore attracted sustained attention, and convolutional systems trained on large retinal corpora now achieve specialist-level discrimination for referable disease [3], [4]. These systems share a structural limitation: they return a scalar severity estimate with no account of the retinal findings that produced it. In a clinical workflow this is consequential. A grader cannot audit a label, a trainee cannot learn from it, and a regulator cannot inspect the basis of a recommendation. Post-hoc saliency methods are frequently offered as a remedy, but their reliability as explanations of medical model behaviour has been forcefully questioned [5].

Vision-language models suggest a different route. A model that emits text can, in principle, name the lesions it observes, connect them to a severity criterion, and propose a follow-up interval—producing an artefact a clinician can read and contest rather than a number they must trust. Instruction-tuned VLMs have been adapted to biomedical domains with encouraging results [6], [7], yet the prevailing assumption is that such adaptation demands multiple high-memory accelerators. That assumption has a distributional consequence: it concentrates capability in well-funded institutions and excludes precisely those low-resource settings where screening shortfalls are most acute.

This paper tests the assumption directly. We ask whether a compact 2B-parameter VLM can be adapted for explainable DR grading within the memory budget of a consumer laptop GPU, and what such a system can and cannot deliver. Our contributions are:

- 1) IDRiD-VQA, a 2,064-pair visual-question-answering corpus for explainable DR assessment, generated

from expert labels rather than pixels so that every reference answer is consistent with ground truth by construction (Section III).

- 2) A memory-frugal adaptation recipe that fits training of Qwen2-VL-2B-Instruct into 3.52 GB of VRAM and approximately 90 minutes on a single 6 GB GPU (Section IV).
- 3) An empirical finding of practical consequence: for this task input resolution dominates the conventional levers of epoch count and optimiser schedule, contributing 11.6 accuracy points at essentially no memory cost (Section VI).
- 4) A critical appraisal of the resulting system, including an error analysis showing that residual failures concentrate in proliferative disease, and an argument that the label-conditioned construction of our corpus bounds what the explanations can be claimed to demonstrate (Section VII).

II. Related Work

A. Automated Diabetic Retinopathy Grading

Deep convolutional classifiers established the modern baseline for DR grading. Gulshan et al. [3] demonstrated ophthalmologist-comparable sensitivity for referable disease on large clinical corpora, and Ting et al. [4] confirmed generalisation across multiethnic screening populations. The IDRiD dataset [8] contributed pixel-level lesion annotation to a subset of images and anchored a public grading challenge whose results remain the standard reference point for this corpus [9]. Subsequent architectural work has exploited the clinical correlation between retinopathy and macular oedema; CANet [10] introduced cross-disease attention and reports the strongest joint grading accuracy on the IDRiD benchmark. All of these systems are discriminative: they optimise an ordinal objective and return a grade, not an account of it.

B. Vision-Language Models in Medicine

General-purpose instruction-tuned VLMs couple a vision encoder to a language decoder and can be steered by natural-language prompts [7]. The Qwen2-VL family [11] is notable for native dynamic resolution, processing images at variable token budgets rather than a fixed grid—a property that proves central to our results. Domain adaptation to biomedicine has been demonstrated by LLaVA-Med [6], which curates biomedical instruction data to specialise a general assistant. Reviews of the field emphasise that the promise of such models in medical imaging remains contingent on rigorous evaluation rather than fluency alone [12].

C. Parameter-Efficient Adaptation

Low-rank adaptation (LoRA) [13] freezes pretrained weights and learns low-rank update matrices, reducing trainable parameters by orders of magnitude. QLoRA [14] composes this with 4-bit NormalFloat quantisation of

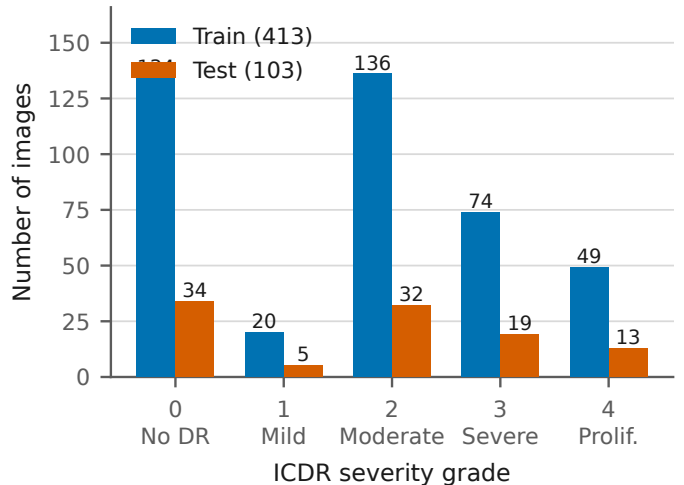


Fig. 1. ICDR grade distribution of the IDRiD disease-grading subset under the canonical 413/103 partition. Mild disease (grade 1) is severely under-represented, with five test images in total.

the frozen backbone, making single-GPU adaptation of large models feasible. Kernel-level implementations such as Unsloth [15] further reduce activation memory through fused operations and custom gradient checkpointing. Our work combines these components, but its contribution is not the components themselves: it is the empirical demonstration that their composition suffices for an explainable clinical task, and the identification of resolution rather than adapter capacity as the binding constraint.

III. Dataset Construction

A. Source Corpus and Split Protocol

We build on the IDRiD disease-grading subset [8], comprising 516 colour fundus photographs at 4288×2848 pixels, each annotated with an ICDR severity grade $g \in \{0, \dots, 4\}$ and a macular-oedema risk label. We adopt the canonical challenge partition of 413 training and 103 testing images [9] to preserve comparability with prior work.

An implementation detail materially affects correctness: the training and testing directories reuse filenames, so an image identifier such as IDRiD_001 is ambiguous across splits. We therefore key every record by the pair (split, identifier) rather than by identifier alone. Keying by filename silently merges the partitions and inflates apparent performance; we report this because the failure mode is easy to introduce and hard to detect downstream.

Fig. 1 shows the resulting grade distribution. The corpus is markedly imbalanced: moderate (grade 2) and absent (grade 0) disease together account for 65% of training images, whereas mild disease (grade 1) is represented by only 20 training and 5 testing images. This skew is a property of the source data, not of our sampling, and it conditions the interpretation of per-class results in Section VI.

B. Label-Grounded Question Answering

Adapting a generative model requires supervision in natural language, which IDRiD does not provide. We synthesise it. For each image we assemble a structured record containing the severity grade, the macular-oedema risk and a vector of lesion-presence indicators for microaneurysms, haemorrhages, hard exudates and soft exudates. Where segmentation masks exist we derive indicators from non-zero pixel support; for the grading-only images we fall back to grade-conditioned clinical priors and record the provenance of every indicator so that inferred and measured evidence remain distinguishable.

These structured records—and not the images—are then supplied to a Gemini Flash model [16], which is prompted to produce four question–answer pairs per case covering (i) grade identification, (ii) lesion identification, (iii) clinical reasoning and (iv) patient recommendation. Conditioning the generator on labels rather than pixels is a deliberate design choice with a specific consequence: because the generator never observes the fundus photograph, it cannot hallucinate findings absent from the ground truth, and every reference answer is consistent with the expert annotation by construction. The cost of this choice is a corresponding limit on explanation diversity, which we address in Section VII.

Generation is batched at eight cases per request and rate-limited; because free-tier quotas are enforced per model, we cycle a pool of Gemini Flash variants as each quota is exhausted, with a deterministic clinical template as a guaranteed fallback. The procedure yields 2,064 pairs (4 per image over all 516 images), of which 1,652 form the training set and 412 the held-out set. In the run reported here every pair was model-generated; no template fallback was required.

IV. Methodology

A. Backbone and Adaptation

We adapt Qwen2-VL-2B-Instruct [11], whose frozen weights are loaded in 4-bit NormalFloat precision following QLoRA [14]. For a frozen projection $W_0 \in \mathbb{R}^{d \times k}$, adaptation learns a low-rank correction realised as the product of two thin matrices,

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} B A x, \quad (1)$$

with $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$ and $r \ll \min(d, k)$. We set $r = \alpha = 16$ and attach adapters to all attention and MLP projections in both the language and vision towers, with no dropout and no bias terms. Only A and B receive gradients, giving 28.9M trainable parameters, or 1.45% of the 2B-parameter backbone.

The objective is the standard autoregressive cross-entropy restricted to answer tokens, so that the model is not rewarded for reproducing the prompt:

TABLE I

Training configuration. The full run completes in approximately 90 minutes on a single laptop RTX 4050 and peaks at 3.52 GB of VRAM.

Component	Setting
Backbone	Qwen2-VL-2B-Instruct, 4-bit NF4
Adapter	LoRA, $r = 16$, $\alpha = 16$, dropout 0
Adapted modules	all attention + MLP (vision & language)
Trainable parameters	28.9M (1.45%)
Input resolution	longer side 768 px ($T_v \approx 486$)
Max sequence length	1,280 tokens
Batch / accumulation	1×8 (effective 8)
Optimiser	8-bit AdamW, linear decay
Learning rate	2×10^{-4}
Epochs / steps	4 / 824
Precision	bfloat16 activations
Wall-clock time	≈ 90 min
Peak VRAM	3.52 GB

$$\mathcal{L} = -\frac{1}{|\mathcal{Y}|} \sum_{t \in \mathcal{Y}} \log p_{\theta}(y_t \mid y_{<t}, \mathbf{v}, q), \quad (2)$$

where $\mathbf{v}$ denotes the visual token sequence, q the question, and $\mathcal{Y}$ the index set of answer tokens.

B. Resolution and the Visual Token Budget

Qwen2-VL encodes images at native aspect ratio into patches of side $p = 14$, merging $m \times m$ neighbourhoods ($m = 2$) before the language decoder. The visual token count for an image of height H and width W is therefore approximately

$$T_v \approx \frac{1}{m^2} \left\lceil \frac{H}{p} \right\rceil \left\lceil \frac{W}{p} \right\rceil. \quad (3)$$

Equation (3) makes the resolution trade-off explicit and quantitative. Resizing a fundus photograph so that its longer side is 512 pixels yields $T_v \approx 216$ tokens; 768 pixels yields 486; 1024 pixels yields 888. Because clinically decisive lesions differ by orders of magnitude in scale—a microaneurysm subtends only a few pixels of the original 4288-pixel image, whereas a cotton-wool spot subtends hundreds—the choice of T_v determines which findings survive downsampling at all. We treat resolution as a first-class hyperparameter and ablate it in Section VI.

C. Fitting Training into 6 GB

Table I lists the configuration. Memory is controlled by four compounding measures: 4-bit backbone quantisation, Unsloth gradient checkpointing [15], a per-device batch size of one with gradient accumulation of eight (preserving an effective batch of eight), and bfloat16 activations throughout—no tensor is materialised in fp32. Optimiser state is held in 8-bit AdamW, which matters because adapter gradients, not frozen weights, dominate the optimiser footprint once the backbone is quantised.

V. Experiments

A. Protocol

All evaluation is performed on the 103 held-out images, which are disjoint from training by construction. Decoding is greedy throughout, so reported figures are deterministic and reproducible. For each test image the model is queried independently for a severity grade and for a clinical rationale; an integer grade is recovered from the free-text response by pattern matching, and we report the number of unparseable responses as a diagnostic. No response failed to parse in any reported run.

B. Metrics

Grading is assessed by exact-match accuracy and by quadratic weighted kappa [17], the conventional measure for ordinal DR grading:

$$\kappa = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}, \quad w_{ij} = \frac{(i-j)^2}{(N-1)^2}, \quad (4)$$

with O the observed confusion matrix, E its expectation under independent marginals and $N = 5$. The quadratic weighting is the reason κ is preferable to accuracy here: confusing absent with proliferative disease is penalised sixteen times more heavily than an adjacent-grade error, which matches the clinical asymmetry of the underlying decision. We interpret magnitudes using the conventional agreement bands of Landis and Koch [18].

Explanation quality is measured by ROUGE-L [19] for lexical overlap and BERTScore [20] for semantic similarity against the reference rationale. We additionally report mean per-image latency and marginal monetary cost.

C. Baselines and Ablation

We compare the adapted model against the unmodified Qwen2-VL-2B backbone evaluated zero-shot under an identical prompt and decoding configuration, isolating the effect of adaptation. To separate the contribution of resolution from that of adaptation, we additionally train an otherwise identical model at 512-pixel input. An API baseline is supported by our evaluation harness but was not executed, as no commercial key was available; we state this explicitly rather than omit the comparison silently.

VI. Results

A. Effect of Adaptation

Table II reports the principal comparison. Adaptation raises exact-match accuracy from 31.1% to 66.0% and quadratic weighted kappa from 0.000 to 0.763. The kappa figure is the more informative of the two. A value indistinguishable from zero indicates that the zero-shot model’s grade predictions carry no ordinal information about the true severity: inspection of its outputs shows verbose, generic descriptions that gravitate toward the majority “moderate” category, so its 31.1% accuracy reflects the base rate of that class rather than any discriminative

TABLE II

Grading and explanation quality on the 103-image IDRid test split. Accuracy is exact match over grades 0–4; κ is quadratic weighted kappa. Latency is mean wall-clock time per image on an RTX 4050; marginal cost is nil for locally executed models.

Model	Acc.	κ	R-L	BScore	s/img
Qwen2-VL-2B (zero-shot)	31.1	0.000	0.081	0.000	10.6
+ QLoRA, 512 px	54.4	0.622	0.235	0.287	4.8
+ QLoRA, 768 px (ours)	66.0	0.763	0.253	0.309	5.6

capability. The adapted model instead attains 0.763, placing it in the upper part of the “substantial agreement” band [18].

Explanation quality improves concurrently: ROUGE-L rises from 0.081 to 0.253 and BERTScore from 0.000 to 0.309. Mean latency falls from 10.6 to 5.6 seconds per image, a 47% reduction, because the adapted model terminates at a structured answer whereas the baseline continues to the token limit.

B. Resolution Ablation

Fig. 2 isolates the effect of input resolution with all other factors held fixed. Increasing the longer side from 512 to 768 pixels—raising T_v from approximately 216 to 486 tokens by Equation (3)—improves accuracy by 11.6 points (54.4% to 66.0%) and kappa by 0.14 (0.622 to 0.763). Explanation metrics improve more modestly, which is consistent with the interpretation that resolution primarily affects what the vision encoder can resolve rather than how the decoder verbalises it.

Two aspects of this result deserve emphasis. First, the gain is large relative to what is typically obtained by tuning epochs or learning rate on a corpus of this size, which suggests that resolution was the binding constraint rather than optimisation. Second, the cost is negligible: peak memory was 3.52 GB at both settings, because gradient checkpointing recomputes activations rather than retaining them, so the additional 270 visual tokens inflate compute and latency but not the memory ceiling. Practitioners operating under a fixed VRAM budget should therefore treat resolution as the first hyperparameter to spend on, not the last.

C. Error Structure

Aggregate metrics conceal the behaviour that matters clinically, so we examine the confusion matrix (Fig. 3) and the per-class breakdown (Table III) directly. Two observations follow, the second of which qualifies the headline result substantially.

First, the ordinal structure of the errors is favourable. Of 103 test images, 66.0% are graded exactly, a further 18.4% fall within one grade, and 15.5% err by two grades or more; 84.5% therefore land within one severity step of the reference. This concentration near the diagonal is what Equation (4) rewards and explains why κ reaches 0.763 despite a much lower exact-match rate.

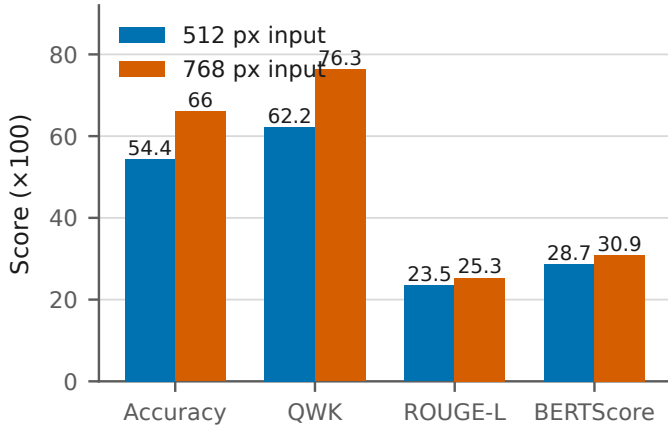


Fig. 2. Input-resolution ablation with all other settings fixed. Raising the longer side from 512 to 768 pixels yields +11.6 accuracy points and +0.14 kappa at unchanged peak memory. Scores are shown $\times 100$ so that a single axis serves all four metrics.

TABLE III

Per-class performance of the adapted model. Recall collapses on minority grades; grade 1 is never predicted. Support is the number of test images per class.

Grade	Supp.	Pred.	Rec. (%)
0 – No DR	34	45	97.1
1 – Mild NPDR	5	0	0.0
2 – Moderate NPDR	32	42	78.1
3 – Severe NPDR	19	9	31.6
4 – Proliferative	13	7	30.8
Overall	103	103	66.0

Second, and more importantly, the model exhibits pronounced majority-class collapse. The predicted-label distribution is (45, 0, 42, 9, 7) for grades 0 through 4: the model assigns grade 0 or grade 2 to 84% of all images, and it never once predicts grade 1 across the entire test set. Per-class recall consequently ranges from 97.1% on grade 0 to 0.0% on grade 1, with severe (31.6%) and proliferative (30.8%) disease also poorly recovered—each predominantly misassigned to grade 2. The aggregate accuracy of 66.0% is thus carried by the two majority classes, which together constitute 64% of the test partition. We regard this, rather than the headline accuracy, as the principal finding of the error analysis, and we examine its causes in Section VII.

D. Comparison with Prior Work

Contextualising these figures requires care about what is being measured. The strongest published results on the IDRiD grading benchmark report joint accuracy, requiring the retinopathy grade and the macular-oedema grade to be simultaneously correct; CANet [10] attains 65.1% under that definition and the leading challenge entry 63.1% [9]. Our 66.0% is a retinopathy-only figure and is therefore not directly comparable: the joint criterion is strictly harder, and specialised discriminative classifiers

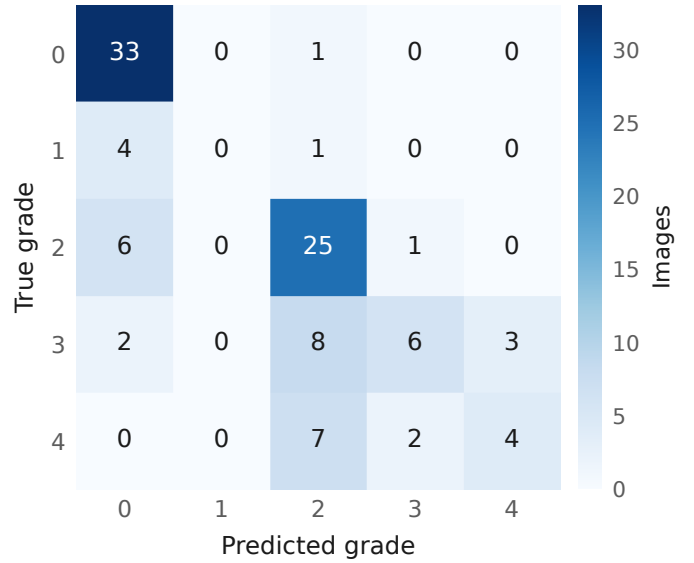


Fig. 3. Confusion matrix of the adapted model on the 103-image test split. Mass concentrates on and adjacent to the diagonal, but the grade-1 column is entirely empty—mild retinopathy is never predicted—and severe and proliferative disease are predominantly misassigned to grade 2.

evaluated on retinopathy alone—particularly those trained on corpora one to two orders of magnitude larger—remain ahead of the system reported here. We claim only that a 2B-parameter generative model adapted on 413 images reaches the same broad performance regime as purpose-built classifiers on this corpus, while additionally producing an auditable rationale and running offline on commodity hardware.

VII. Discussion

A. Two Causes of Minority-Class Collapse

The failure pattern in Table III has two distinct causes, and separating them matters because they demand different remedies.

The first is data scarcity acting through an imbalanced objective. Grade 1 is represented by 20 training images out of 413 (Fig. 1). Under a token-level cross-entropy objective (Equation (2)) with no class reweighting, the loss is minimised almost as effectively by never emitting the token sequence for “mild” as by learning its visual signature, since the expected penalty for abandoning a class of prior probability 0.048 is small. The model duly learns this degenerate solution: grade 1 is predicted zero times. This is a textbook consequence of optimising an unweighted likelihood on a skewed corpus, and it is not remedied by higher resolution—the 512-pixel model failed on grade 1 identically. Class-balanced sampling, loss reweighting, or explicit oversampling of rare grades are the appropriate interventions, and we did not apply them.

The second cause is supervision that does not encode the decision boundary, and it explains the specific confusion of grades 3 and 4 with grade 2. Lesion indicators for

grading-only images derive from grade-conditioned priors, and those priors assign severe and proliferative disease near-identical lesion vectors, because both present microaneurysms, haemorrhages and exudates. The feature that clinically separates them—retinal neovascularisation—is absent from the indicator vector entirely. The training rationales for the two classes are therefore close to interchangeable, and the model is asked to discriminate a boundary its supervision never represents. This is a limitation of dataset construction rather than of the backbone, and it admits a concrete remedy: propagating the pixel-level neovascularisation evidence available in the IDRiD segmentation subset into the indicator vector, at the cost of restricting that evidence to the 81 images for which masks exist.

We emphasise that neither cause is exposed by aggregate accuracy or by κ . A system reporting 66.0% accuracy and “substantial” ordinal agreement is, on inspection, one that cannot detect mild disease at all—precisely the stage at which screening intervention is most valuable. This dissociation between headline metrics and clinical utility is, in our view, the most transferable lesson of the present study.

B. The Limits of These Explanations

A second consequence of label-conditioned generation warrants explicit statement, because it bounds the claims this paper can support. Our reference rationales are derived from expert annotations, so they are factually consistent with ground truth; but they are not evidence that the model’s internal computation follows the reasoning it articulates. A fluent, correct-sounding rationale accompanying a correct grade does not establish that the rationale caused the grade. This is the failure mode that Ghassemi et al. [5] identify in medical explainability generally, and our architecture does not escape it. A related weakness is mechanical: grade and rationale are obtained from independent queries, so the two can disagree—we observed instances in which a correct grade accompanied a rationale describing a different severity. Joint decoding of a single structured report would remove this inconsistency and is the most immediate improvement we can identify.

C. Practical Implications

Subject to those caveats, the engineering result is clear. Adaptation of an explainable DR grader required approximately 90 minutes and 3.52 GB of VRAM, which is within the envelope of consumer hardware widely available to individual researchers, teaching hospitals and screening programmes in resource-constrained settings. Inference is fully local, incurring no per-image cost and no transmission of patient imagery to third-party infrastructure—a property with governance value independent of accuracy. The finding that resolution dominates is additionally actionable: it identifies the cheapest available lever for practitioners who cannot enlarge their corpus.

D. Threats to Validity

Our evidence base is a single-centre corpus of 516 images captured on one camera system; generalisation across devices, populations and image quality regimes is untested and should not be assumed. The test partition contains only five grade 1 images, so per-class estimates for mild disease carry wide and unreported uncertainty. We report a single training run rather than a distribution over seeds, so the ablation in Fig. 2 should be read as a strong indication rather than a variance-controlled measurement. Finally, no clinician assessed the faithfulness or usefulness of the generated rationales; automatic overlap metrics such as ROUGE-L and BERTScore measure similarity to a reference text, which is a weak proxy for clinical adequacy.

VIII. Conclusion

We have shown that explainable diabetic retinopathy grading can be adapted end-to-end within the memory budget of a single 6 GB consumer GPU. Combining a compact vision-language backbone, a label-grounded question-answering corpus and 4-bit low-rank adaptation, the resulting system raises exact-match accuracy from 31.1% to 66.0% and quadratic weighted kappa from 0.000 to 0.763 on the canonical IDRiD test split, while emitting a structured clinical rationale and operating offline at no marginal cost. Our ablation identifies input resolution as the dominant factor for this task, contributing 11.6 accuracy points at unchanged peak memory—a result we expect to transfer to other tasks whose decisive image features are small relative to the frame.

We are deliberate about what has not been shown, because the aggregate figures above are not the whole result. Per-class analysis reveals that the system collapses onto the majority grades: it never predicts mild retinopathy and recovers severe and proliferative disease in fewer than a third of cases. A model that cannot detect early disease is not a screening instrument, whatever its kappa. We trace this to two separable causes—an unweighted objective over a corpus in which mild disease is 4.8% of training images, and lesion indicators that do not encode the neovascularisation boundary—and note that neither is visible in accuracy or κ alone. Its rationales, similarly, are consistent with ground truth but not demonstrated to be faithful to its own computation, and its evidence base is a single-centre corpus of 516 images.

The natural continuations follow from these limits rather than from the results: class-balanced or reweighted training to recover the minority grades, propagation of pixel-level neovascularisation evidence into supervision, joint decoding of grade and rationale so the two cannot diverge, extension across multiple public corpora to test cross-device generalisation, and—indispensably before any claim of clinical utility—a reader study in which ophthalmologists assess whether the generated rationales are faithful and actionable. We release the dataset, adapter

weights and full training and evaluation code to make each of these steps directly reproducible.¹

Ethics Statement

This work is a research prototype and is not a medical device. Its outputs must not be used for diagnosis, triage or treatment decisions. The study uses the publicly available IDRiD dataset [8] under its licence and involves no new patient recruitment and no personally identifying information. Reference rationales are model-generated and may contain clinical errors despite being conditioned on expert labels. We note that a system which produces fluent clinical language can invite unwarranted confidence, and that this risk is heightened rather than reduced when the language is well-formed; deployment would require prospective validation and human oversight.

Reproducibility Statement

All results derive from a single publicly available dataset and a released codebase. Decoding is greedy and seeds are fixed, so reported metrics are deterministic given the released adapter. The training configuration is given in full in Table I, and the exact package versions required—several of which are load-bearing, as the adaptation stack is sensitive to version drift—are pinned in the repository lockfile.

References

- [1] Z. L. Teo, Y.-C. Tham, M. Yu, M. L. Chee, T. H. Rim, N. Cheung, M. M. Bikbov, Y. X. Wang, Y. Tang, Y. Lu et al., “Global prevalence of diabetic retinopathy and projection of burden through 2045: Systematic review and meta-analysis,” *Ophthalmology*, vol. 128, no. 11, pp. 1580–1591, 2021.
- [2] C. Wilkinson, F. L. Ferris III, R. E. Klein, P. P. Lee, C. D. Agardh, M. Davis, D. Dills, A. Kampik, R. Pararajasegaram, and J. T. Verdager, “Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales,” *Ophthalmology*, vol. 110, no. 9, pp. 1677–1682, 2003.
- [3] V. Gulshan, L. Peng, M. Coram, M. C. Stumpe, D. Wu, A. Narayanaswamy, S. Venugopalan, K. Widner, T. Madams, J. Cuadros et al., “Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs,” *JAMA*, vol. 316, no. 22, pp. 2402–2410, 2016.
- [4] D. S. W. Ting, C. Y.-L. Cheung, G. Lim, G. S. W. Tan, N. D. Quang, A. Gan, H. Hamzah, R. Garcia-Franco, I. Y. San Yeo, S. Y. Lee et al., “Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes,” *JAMA*, vol. 318, no. 22, pp. 2211–2223, 2017.
- [5] M. Ghassemi, L. Oakden-Rayner, and A. L. Beam, “The false hope of current approaches to explainable artificial intelligence in health care,” *The Lancet Digital Health*, vol. 3, no. 11, pp. e745–e750, 2021.
- [6] C. Li, C. Wong, S. Zhang, N. Usuyama, H. Liu, J. Yang, T. Naumann, H. Poon, and J. Gao, “Llava-med: Training a large language-and-vision assistant for biomedicine in one day,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [7] H. Liu, C. Li, Q. Wu, and Y. J. Lee, “Visual instruction tuning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [8] P. Porwal, S. Pachade, R. Kamble, M. Kokare, G. Deshmukh, V. Sahasrabudhe, and F. Meriaudeau, “Indian diabetic retinopathy image dataset (idrid): A database for diabetic retinopathy screening research,” *Data*, vol. 3, no. 3, p. 25, 2018.
- [9] P. Porwal, S. Pachade, M. Kokare, G. Deshmukh, J. Son, W. Bae, L. Liu, J. Wang, X. Liu, L. Gao et al., “Idrid: Diabetic retinopathy – segmentation and grading challenge,” *Medical Image Analysis*, vol. 59, p. 101561, 2020.
- [10] X. Li, X. Hu, L. Yu, L. Zhu, C.-W. Fu, and P.-A. Heng, “Canet: Cross-disease attention network for joint diabetic retinopathy and diabetic macular edema grading,” *IEEE Transactions on Medical Imaging*, vol. 39, no. 5, pp. 1483–1493, 2020.
- [11] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, K. Chen, X. Liu, J. Wang, W. Ge, Y. Fan, K. Dang, M. Du, X. Ren, R. Men, D. Liu, C. Zhou, J. Zhou, and J. Lin, “Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution,” *arXiv preprint arXiv:2409.12191*, 2024.
- [12] P. Rajpurkar and M. P. Lungren, “The current and future state of ai interpretation of medical images,” *New England Journal of Medicine*, vol. 388, no. 21, pp. 1981–1990, 2023.
- [13] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “Lora: Low-rank adaptation of large language models,” in *International Conference on Learning Representations (ICLR)*, 2022.
- [14] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “Qlora: Efficient finetuning of quantized llms,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [15] D. Han, M. Han, and Unsloth Team, “Unsloth: Fast and memory-efficient llm fine-tuning,” <https://github.com/unslothai/unsloth>, 2024.
- [16] Gemini Team, Google, “Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context,” *arXiv preprint arXiv:2403.05530*, 2024.
- [17] J. Cohen, “Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit,” *Psychological Bulletin*, vol. 70, no. 4, pp. 213–220, 1968.
- [18] J. R. Landis and G. G. Koch, “The measurement of observer agreement for categorical data,” *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.
- [19] C.-Y. Lin, “Rouge: A package for automatic evaluation of summaries,” in *Text Summarization Branches Out*, 2004, pp. 74–81.
- [20] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “Bertscore: Evaluating text generation with bert,” in *International Conference on Learning Representations (ICLR)*, 2020.

¹Code: <https://github.com/vermayuvraj/idrid-vqa>. Model: <https://huggingface.co/vermayuvraj/idrid-qwen2vl-2b-qlora>. Dataset: <https://huggingface.co/datasets/vermayuvraj/idrid-vqa>.